

ER Diagram

Introduction:- The Entity-Relationship (ER) diagram is a conceptual data modeling technique used to visually represent the structure of a database system. It shows how data is stored, the entities involved & the relationships between them. ER diagram helps software engineers understand system requirements & design efficient database structures before implementation. They are widely used during the analysis phase of the software development life cycle (SDLC).

Component & Features of ER Diagram:-

(a) Component

i) Entity $\rightarrow$ A real world object or concept that has an independent existence.

Ex - Student, Course, Employee, Department.

ii) Attributes $\rightarrow$ Characteristics or properties of an entity.

Ex - Student name, Roll No, Course ID.

iii) Relationship $\rightarrow$ Association between two or more entities.

Ex - Student enrolls in courses.

iv) Primary Key $\rightarrow$ Attributes that uniquely identifies each entity instance.

v) Cardinality $\rightarrow$ Specifies the numerical mapping between entities.

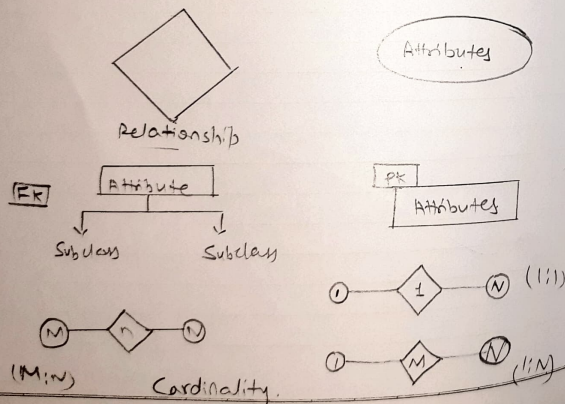
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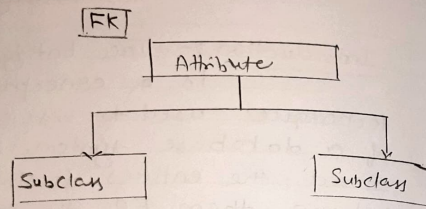
One-to-One (1:1)

One-to-Many (1:M)

Many-to-Many (M:N)

Teacher's Signature





Derived
Attributes

(vi) Foreign key $\rightarrow$ Attribute that links to primary key (PK) of another table.
used to maintain relationship between tables.
ensure referential integrity in database.

(vii) Weak entity $\rightarrow$ Entity that depend on another entity.

Ex Order-Item depends on order.

(ix) Derived Attributes $\rightarrow$ Attributes whose value can be calculated from another attributes.

Ex Age derived from Date of Birth.

Example of ER Diagram

A typical diagram for this example would show:

- Student & Course connected through Enrollment (M:N $\rightarrow$ resolve using a relationship entity).
- Instructor connected to Course with a 1:N relationship.
- Cardinality symbols near each connector.
- Primary Key underlined.

Viva Question

Q1. What is an ER diagram and why is it used?
 Ans. ER (Entity - Relationship) diagram is a visual representation of entities, attributes & relationship in a database system. It is used to model the structure of a database, identify how data is connected and help in designing the database during the analysis phase of software development.

Q2. Differentiate between an entity & attribute?
 Ans.

<u>Entity</u>	<u>Attribute</u>
A real-world object or concept that has an independent existence. Ex Student, Course, Employee.	A property of characteristic the describes an entity. Ex Student Name, Course ID, Employee salary.

Q3. What is the Purpose of a primary key in ER diagram?
 Ans. A primary key (PK) is used to uniquely identify each instance (record) of an entity. It ensure that no two records in a table have the same value for that attribute & helps maintain data integrity.

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Q4 Explain the concept of Cardinality with example.

Ans Cardinality defines the numerical relationship between two entities - how many instances of one Entity relate to instances of another.

Types

- One-to-one (1:1) → One person has one passport.
- One-to-many (1:N) → One teacher teaches many student.
- Many-to-many (M:N) → Students enroll in many courses. & many courses have many students.

Q5 What the difference between a weak entity & a strong entity.

Strong entity

Weak entity

- | | |
|---------------------------|---|
| • Has its own primary key | • Doesn't have a primary key of its own. |
| • Exists independently | • Depends on a strong entity for its existence. |
| • Ex Student, Courses | • Ex Order-item depends on Order. |